

CS341 Home work Problems

1. Suppose there is exactly one packet switch between a sending host and a receiving host. The transmission rates between the sending host and the switch and between the switch and the receiving host are R_1 and R_2 , respectively. Assuming that the switch uses store-and-forward packet switching, what is the total end-to-end delay to send a packet of length L ? (Ignore queuing, propagation delay, and processing delay.)
2. Hosts A and B are each connected to a switch S via 10-Mbps links as in Figure below. The propagation delay on each link is $20\ \mu\text{s}$. S is a store-and-forward device; it begins retransmitting a received packet $35\ \mu\text{s}$ after it has finished receiving it. Calculate the total time required to transmit 10,000 bits from A to B.



- a. As a single packet.
 - b. As two 5,000-bit packets sent one right after the other.
3. Suppose Host A wants to send a large file to Host B. The path from Host A to Host B has three links, of rates $R_1 = 500\text{kbps}$, $R_2 = 2\text{Mbps}$, and $R_3 = 1\text{Mbps}$.
 - a. Assuming no other traffic in the network, what is the throughput for the file transfer?
 - b. Suppose the file is 4 million bytes. Dividing the file size by the throughput, roughly how long will it take to transfer the file to Host B?
 - c. Repeat (a) and (b), but now with R_2 reduced to 100 kbps.
 4. Consider a point-to-point link 2 km in length. At what bandwidth would propagation delay (at a speed of $2 \times 10^8\text{m/sec}$) equal transmit delay for 100-byte packets? What about 512-byte packets?
 5. Calculate the latency (from first bit sent to last bit received) for:
 - a. A 1-Gbps Ethernet with a single store-and-forward switch in the path, and a packet size of 5,000 bits. Assume that each link introduces a propagation delay of $10\ \mu\text{s}$ and that the switch begins retransmitting immediately after it has finished receiving the packet.
 - b. Same as (a) but with three switches.
 - c. Same as (b) but assume the switch implements cut-through switching: it is able to begin retransmitting the packet after the first 128 bits have been received.
 6. Suppose a 100-Mbps point-to-point link is being set up between Earth and a new lunar colony. The distance from the moon to Earth is approximately 385,000 km, and data travels over the link at the speed of light— $3 \times 10^8\text{ m/s}$.
 - a. Calculate the minimum RTT for the link.

- b. Using the RTT as the delay, calculate the delay $\times$ bandwidth product for the link.
- c. A camera on the lunar base takes pictures of Earth and saves them in digital format to disk. Suppose Mission Control on Earth wishes to download the most current image, which is 25 MB. What is the minimum amount of time that will elapse between when the request for the data goes out and the transfer is finished?
